MACHINE LEARNING COURSE

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The tasks are relevant to the dataset "**Boston Housing**"

The data includes 17 features with target feature - "MEDV"

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	MEDV
0	0.00632	18.0	2.31	0.0	0.538	6.575	65.2	4.0900	1.0	296.0	15.3	396.90	4.98	24.0
1	0.02731	0.0	7.07	0.0	0.469	6.421	78.9	4.9671	2.0	242.0	17.8	396.90	9.14	21.6
2	0.02729	0.0	7.07	0.0	0.469	7.185	61.1	4.9671	2.0	242.0	17.8	392.83	4.03	34.7
3	0.03237	0.0	2.18	0.0	0.458	6.998	45.8	6.0622	3.0	222.0	18.7	394.63	2.94	33.4
4	0.06905	0.0	2.18	0.0	0.458	7.147	54.2	6.0622	3.0	222.0	18.7	396.90	5.33	36.2

Question 1 (Data preparation)

a. Using Correlation Matrix to display correlations between the indicated variables (should be visualised by heatmap).

b. From the 1.a result, find two features which is related the most and linear with **MEDV**

Question 2 (Linear regression model)

a. Split the data into **d.train** training set and **d.test** test set 80:20.

b. Perform a linear regression using two features to predict **MEDV**

c. Evaluation the model with the metric RMSE